Final Examination Practice Sheet

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Section(s): K3P25RB

Syllabus Covered: Unit 1 to Unit 6

Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)

Purpose: self-practice for the end-term examination

Instructions: attempt every question without notes, then check your answers with your instructor. For the long questions, write structured answers in the form you would use in the examination.

Unit 1: Foundations of Artificial Intelligence

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. Which test judges a machine by whether an interrogator, chatting by text, can reliably tell it apart from a human?

(A) The Chinese Room argument
(B) The Turing Test
(C) The Lighthill review
(D) The Winograd schema

2. A voice assistant can set alarms very well but cannot learn to play chess or transfer its skill to any other task. Which category of AI is it?

(A) General AI
(B) Super AI
(C) Self-aware AI
(D) Narrow AI

3. Which component of a pick-and-place robot acts upon its environment?

(A) The gripper and motors (actuators)
(B) The camera (sensor)
(C) The percept sequence
(D) The performance measure

4. In a game of chess the agent can see the whole board at every move. What property does this task environment have?

(A) Partially observable
(B) Stochastic
(C) Fully observable
(D) Continuous

5. A loan model trained on past approval decisions consistently gives lower scores to applicants from one district. Which concern does this illustrate?

(A) Overfitting
(B) Data leakage
(C) Bias inherited from the historical training data
(D) Underfitting

6. A spam filter classifies each incoming email on its own, and one decision does not affect the next. What kind of task environment is this?

(A) Episodic
(B) Sequential
(C) Dynamic
(D) Multi-agent

7. Six months after a fruit-grading model was deployed, its accuracy has dropped although its code is unchanged. What is the most likely explanation?

(A) A bug was introduced
(B) The model is overfitting its code
(C) A hardware fault
(D) The incoming data has drifted, so the model needs monitoring and retraining

8. Which statement about machine learning and deep learning is correct?

(A) Machine learning is a subset of deep learning
(B) Deep learning is a subset of machine learning that uses multi-layer neural networks
(C) They are unrelated fields
(D) Deep learning does not need data

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Explain the PEAS framework, and apply it to an autonomous lawn-mowing robot working on a university campus. (10 Marks)

(a) Explain what each of the four PEAS elements - performance measure, environment, actuators and sensors - is meant to capture. *[4]*

(b) Write out what each of the four would contain for this robot. *[4]*

(c) State whether the campus lawn is fully or partially observable, with one reason. *[2]*

10. Distinguish between Artificial Intelligence, Machine Learning and Deep Learning, using healthcare as the setting. (10 Marks)

(a) Describe the scope of each of the three and state how they are related to one another. *[6]*

(b) Give one healthcare application of each, and state in one sentence why it belongs where you placed it. *[4]*

Unit 2: Data Preparation and Machine Learning Workflow

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. Sensor readings range from 20 to 120. After min-max normalisation to the range 0 to 1, what does a reading of 50 become?

(A) 0.30 (B) 0.50

(C) 0.25 (D) 0.42

2. Heights in a class have mean 70 and standard deviation 5. Aman's height is 82.5. What is his z-score?

(A) -2.5 (B) 12.5

(C) 2.5 (D) 1.5

3. A feature has first quartile 30 and third quartile 50. Using the 1.5 x IQR rule, is the reading 85 an outlier?

(A) No, as the upper fence is 90 (B) Yes, as the upper fence is 80

(C) No, as the upper fence is 85 (D) Yes, simply because it is above Q3

4. 4-fold cross validation is applied to a dataset of 1,000 rows. How many rows are used for training in each fold?

(A) 250 (B) 1,000

(C) 800 (D) 750

5. A NumPy array holds 4 images, each 3 pixels high and 2 pixels wide, so its shape is (4, 3, 2). The values are summed over the height axis (axis=1). What is the shape of the result?

(A) (4, 3) (B) (4, 2)

(C) (3, 2) (D) (24,)

6. An analyst fills missing values with the mean of the entire dataset and only then splits it into training and test sets. What is the problem?

(A) There is no problem (B) It causes underfitting

(C) It causes class imbalance (D) Information from the test set leaks into training, so the test score is optimistic

7. A feature "city" holds Delhi, Pune or Chennai, with no natural order. Which encoding is appropriate for a linear model?

(A) One-hot encoding, since the categories have no order (B) Ordinal encoding 1, 2, 3

(C) Drop the column (D) Encode by alphabetical rank

8. Which Pandas method reports the non-null count, data type and memory usage of every column?

(A) describe() (B) head()

(C) info() (D) shape

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply a preprocessing pipeline to a hospital table that holds a blood-pressure column in which 5 per cent of the entries are missing and a few readings are obvious typing errors (such as 1,200), a categorical blood-group column with eight values, and an annual-income column ranging from 0 to 50 lakh rupees. (10 Marks)

(a) State the treatment you would apply to each of the three columns, with a one-line reason for each. *[6]*

(b) State whether scaling should be carried out before or after the train-test split, and give the reason. *[4]*

10. Examine and correct the following setup: an analyst splits a dataset 70 to 30, tunes a model by checking it repeatedly on the 30 per cent part until accuracy stops improving, and reports that accuracy as the expected performance on new data. (10 Marks)

(a) Identify the methodological error and explain what it does to the reported figure. *[5]*

(b) Describe the corrected workflow, using cross validation on the training partition. *[5]*

Unit 3: Supervised Machine Learning

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A model predicts salary (in thousands of rupees) as 3.2 + 1.5 x years of experience. What salary does it predict for someone with 8 years of experience?

(A) 12.0 (B) 11.2

(C) 13.7 (D) 15.2

2. A regression line has slope 2 and passes through the point (5, 17). What is its intercept?

(A) 5 (B) 7

(C) 12 (D) 17

3. A logistic regression model gives a linear score of +0.8 for an observation. With a threshold of 0.5, which class is predicted?

(A) Class 0, since 0.8 is less than 1 (B) Class 0, since the sigmoid gives 0.31

(C) Class 1, since sigmoid(0.8) is about 0.69, above 0.5 (D) It cannot be decided

4. A 5-Nearest Neighbour classifier finds neighbours labelled Pass, Fail, Pass, Pass and Fail. Which label is predicted by majority vote?

(A) Pass (B) Fail

(C) A tie (D) Undefined

5. In a two-class Naive Bayes problem the unnormalised score for class A is 0.03 and for class B is 0.09. What posterior probability is assigned to class B?

(A) 0.75 (B) 0.25

(C) 0.09 (D) 0.33

6. A decision tree scores 98 per cent on training data but 60 per cent on test data. What does this indicate?

(A) Underfitting (B) Data leakage

(C) Overfitting; prune or limit its depth (D) A well-fitted model

7. Why is the Naive Bayes classifier called naive?

(A) It uses very few features (B) It assumes the features are independent of one another given the class

(C) It works only with binary data (D) It uses no probabilities

8. A student scales her features and her K-Nearest Neighbour accuracy improves, yet scaling makes no difference at all to her decision tree. Why?

(A) Trees use distances (B) KNN ignores units

(C) Both models need scaling equally (D) KNN uses distances, so wide-range features dominate; a tree splits on one feature at a time by threshold

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply the bias-variance framework to three models trained on the same dataset. Model X has training error 30 per cent and test error 31 per cent. Model Y has training error 2 per cent and test error 20 per cent. Model Z has training error 8 per cent and test error 10 per cent. (10 Marks)

(a) For each model, state whether it is underfitting, overfitting or reasonably fitted, and justify your answer from its pair of error figures. *[6]*

(b) State one change you would make to Model Y and the effect you expect it to have. *[4]*

10. Compare logistic regression, K-Nearest Neighbour and Random Forest as classifiers for a drone that must decide, from ten sensor readings, whether a landing spot is safe. (10 Marks)

(a) For each of the three, state how it forms its decision boundary. *[6]*

(b) Recommend one of the three for use on board the drone, where a decision is needed within a few milliseconds, and justify the choice. *[4]*

Unit 4: Unsupervised Learning

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A bike shop records 500 transactions, and {helmet, gloves} appears in 115 of them. What is the support of this itemset?

(A) 0.115 (B) 0.23

(C) 0.43 (D) 0.50

2. Item A appears in 250 transactions and A together with B appears in 200. What is the confidence of the rule A -> B?

(A) 0.80 (B) 0.25

(C) 1.25 (D) 0.20

3. Principal components explain 45, 30, 15 and 10 per cent of the variance. How many are needed to reach at least 70 per cent?

(A) 1 (B) 3

(C) 4 (D) 2

4. k-Means with k = 2 is applied to the one-dimensional points 2, 4, 20 and 22. What are the two centroids at convergence?

(A) 2 and 22 (B) 12 and 12

(C) 3 and 21 (D) 4 and 20

5. PCA is applied to 80 features and 20 components are retained. What is the percentage reduction in the number of features?

(A) 75 per cent (B) 25 per cent

(C) 60 per cent (D) 20 per cent

6. The itemset {pen, pencil} has support 0.20. What follows for the support of {pen, pencil, eraser}?

(A) It is exactly 0.20 (B) It is at least 0.20

(C) It is 0.40 (D) It is at most 0.20

7. How does agglomerative hierarchical clustering differ from divisive clustering?

(A) Both merge clusters (B) Divisive merges clusters bottom-up

(C) Agglomerative merges bottom-up; divisive splits top-down (D) Both split clusters

8. What happens to the points that DBSCAN labels as noise?

(A) They form a cluster of their own (B) They lie in sparse regions and are left unassigned

(C) They become centroids (D) They are deleted before clustering

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. A car workshop logs the parts replaced at each service. Of 1,000 service visits, the air filter appears in 300, the oil filter appears in 250, and both appear together in 150. (10 Marks)

(a) Define support, confidence and lift. *[3]*

(b) Compute the support, the confidence and the lift for the rule "if the air filter is replaced then the oil filter is replaced", showing the working. *[5]*

(c) Interpret the value of the lift you obtain. *[2]*

10. Compare k-Means, agglomerative hierarchical clustering and DBSCAN. (10 Marks)

(a) For each of the three, state the cluster shape it assumes and whether the number of clusters must be supplied in advance. *[6]*

(b) A survey drone records LiDAR points that trace curved river banks, together with some isolated spurious readings. Recommend the most suitable of the three and justify the choice. *[4]*

Unit 5: Model Evaluation and Improvement

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A model has 30 true positives and 20 false positives. What is its precision?

(A) 0.40 (B) 0.75

(C) 0.60 (D) 0.30

2. A model has precision 0.8 and recall 0.5. What is its F1 score (to two decimals)?

(A) 0.62 (B) 0.65

(C) 0.40 (D) 0.80

3. 5-fold cross validation is applied to 1,500 rows. How many rows are in the validation part of each fold?

(A) 1,200 (B) 300

(C) 150 (D) 500

4. In a dataset, 99 per cent of records belong to one class. Why is accuracy misleading here?

(A) Accuracy cannot be computed (B) Accuracy is always 50 per cent

(C) It requires a regression model (D) Predicting the majority class alone already scores 99 per cent

5. A cancer-screening team moves its classifier's decision threshold from 0.5 down to 0.25 so that fewer cases are missed. What happens to recall?

(A) It falls (B) It does not change

(C) It becomes zero (D) It rises, because more cases are called positive

6. A model gives every patient who has the disease a higher risk score than every patient who does not. What ROC-AUC does it achieve?

(A) 0.5 (B) 1.0

(C) 0 (D) 0.75

7. How do precision and recall differ?

(A) Precision is measured over the predicted positives; recall over the actual positives (B) Precision is over actual positives; recall over predicted positives

(C) Both are measured over all records (D) Precision is measured over the negatives

8. A model scores 0.93 in cross validation but 0.60 on data collected eight months later. What is the most likely reason?

(A) Too many folds were used (B) The model underfits

(C) The data has drifted over time (D) There is a bug in the metric

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. A model that screens chest X-rays for a disease is evaluated on 2,000 X-rays, of which 100 are genuinely positive. The model flags 150 X-rays as positive, and 80 of those flagged are genuinely positive. (10 Marks)

(a) Construct the confusion matrix. *[3]*

(b) Compute the accuracy, the precision and the recall, showing the working. *[5]*

(c) State which single metric matters most here, given that a missed case is far more serious than an extra review. *[2]*

10. Compare grid search and random search as strategies for hyperparameter tuning. (10 Marks)

(a) State how each of the two explores the search space. *[5]*

(b) Compute the total time needed to search an exhaustive grid of 3 by 4 by 5 values for a gradient-boosting model, where a single fit takes 30 seconds and 4-fold cross validation is used. Show the working. *[5]*

Unit 6: AI for Intelligent Systems and Robotics

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A game agent earns 4 points now, 2 points at the next step and 8 points at the step after. With a discount factor of 0.5, what is this sequence worth, measured from now?

(A) 7 (B) 14

(C) 5 (D) 10

2. A predicted box of area 100 and a ground-truth box of area 80 overlap in a region of area 60. What is the intersection over union?

(A) 0.60 (B) 0.75

(C) 0.80 (D) 0.50

3. Using a bag-of-words representation over the sentences "drone flies high" and "drone lands low", what is the vocabulary size?

(A) 3 (B) 6

(C) 5 (D) 4

4. A detector returns boxes with confidences 0.91, 0.58, 0.66, 0.35 and 0.72. With a threshold of 0.6, how many detections are kept?

(A) 2 (B) 3

(C) 4 (D) 5

5. A robot must react within 120 ms. Its detector takes 50 ms, its planner 40 ms, and running the detector remotely adds a 40 ms network round trip. Does the remote set-up meet the deadline?

(A) Yes, it takes 90 ms (B) No, the total of 130 ms exceeds 120 ms

(C) Yes, 130 ms is within the limit (D) No, the total is 170 ms

6. A system looks at a photo and simply answers "cat", without saying where the cat is. Which vision task is it performing?

(A) Image classification (B) Object detection

(C) Semantic segmentation (D) Object tracking

7. A robot vision model trained only on daylight images fails at night. What is the reason?

(A) It has too many parameters (B) It overfits the labels

(C) Its reward is too small (D) The input distribution has shifted away from the training data

8. A cleaning robot always takes the corridor that worked best so far and never tries the others. What risk does this create?

(A) None, since the best route is always known (B) The robot will explore too much

(C) It may stay stuck on a poor route because better routes are never tried (D) Its sensors will fail

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply the reinforcement learning framework to a drone that must deliver medicines from the campus pharmacy to a hostel while avoiding trees and buildings. (10 Marks)

(a) Define the agent, the environment, the state, the set of actions and the reward function for this task. *[6]*

(b) Explain the trade-off between exploration and exploitation as it applies to this drone. *[4]*

10. Compare image classification with object detection as the vision component of a library robot that must stop for people and trolleys in its aisles. (10 Marks)

(a) State what each of the two produces as its output and what each requires of its training data. *[6]*

(b) A predicted box of area 120 square pixels overlaps a ground-truth box of area 100 square pixels by 66 square pixels. Compute the intersection over union, showing the working, and state whether the detection is accepted at a threshold of 0.5. *[4]*

----- End of Practice Sheet -----